

The SPO portfolio tree: model v1

The model and its training, with the equations; then what the planted-threshold experiment says

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24 September 2026

Outline

1. What we do, and how we test it
2. The model: the tree, the decision problem, the loss, the training
3. Model v1: the split-search options and the frozen configuration
4. The planted-threshold experiment: recovery, performance, trading rate, the options, v1 on 100 datasets, what can be found, v1 against signal, volatility and half-life
5. A fair assessment, and what comes next

What we do, and how we test it

Problem

- weights of n assets every h days, from features x_t , with fees
- a decision problem, not a forecast

Model

- small tree $\rightarrow$ leaf scores $\rightarrow$ mean–variance optimizer with fees
- loss: decision regret through the optimizer
- splits confirmed on the tree's own traded path
- Out of sample Sharpe pruning

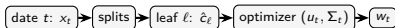
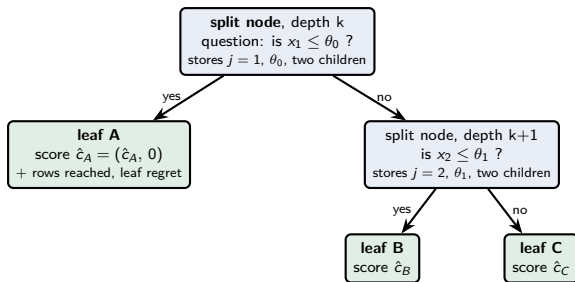
Test

- synthetic market, one feature x , planted threshold d
- benchmark: *ideal rule* = optimizer fed $\mathbb{E}[r \mid x]$
- constant volatility; asset vs cash; weekly unless stated

Model v1

- frozen 24 September 2026
- $\approx 15,000$ fitted trees behind the numbers
- validated on 100 datasets

The tree: what a node stores



- tree = nested questions "is $x_j \leq \theta$?", ending in leaves
- **split node**: feature j , threshold θ , two children; no score
- **leaf**: one score vector $\hat{c}_\ell = \left(\hat{c}_\ell^{(1)}, 0 \right)$ (only the difference matters); rows reached; leaf regret
- holdings u_t , weights w_t , returns: per date, not per node
- depth $\leq 3 \Rightarrow \leq 8$ leaves $\Rightarrow \leq 8$ scores
- growth: leaf $\rightarrow$ split node + 2 leaves; pruning: the reverse
- use: $x_t \rightarrow$ leaf $\rightarrow \hat{c}_\ell \rightarrow$ optimizer $\rightarrow w_t$

The decision: mean–variance with proportional fees

Notation

- u_t : holdings between time $(t-1)^+$ and t^- ,
 $u_{t+1} = \frac{w_t \circ (1 + r_t)}{1 + r_t^\top w_t}$, $u_1 = \mathbf{1}/n$
- c : the tree's scores, in units of the return; cash = 0 (only the score differences matter)
- Σ : covariance of the h -day return = $h \times$ the covariance of the last 180 daily returns; $\sigma_h^2 = h\hat{\sigma}^2$
- f : fee per traded notional, 3 bp on the asset, 0 on cash; $\lambda = 1$: risk aversion, weight of the h -day variance
- $\mathcal{W} = \{w : 0 \leq w_i \leq \bar{w}, \sum_i w_i = 1\}$

$$w_t^*(c; u_t, \Sigma) = \arg \max_{w \in \mathcal{W}} \underbrace{c^\top w}_{\text{gain}} - \underbrace{f^\top |w - u_t|}_{\text{fees}} - \underbrace{\lambda w^\top \Sigma w}_{\text{risk}}$$

Properties

- strictly concave $\Rightarrow$ unique w^*
- $w^*(c + \alpha \mathbf{1}) = w^*(c)$: only score differences matter
- exact solution: 5 states per weight (0, cap = $\bar{w}$, u_i , buy, sell) $\Rightarrow 5^n$ linear systems, best feasible one

Asset against cash: the position window and the no-trade band

One risky asset (weight w , score c , weight held before the decision u) against cash (score 0, no fee, no variance); $\kappa = 2\lambda\sigma_h^2$:

$$w^* = \arg \max_{0 \leq w \leq 1} c w - f |w - u| - \frac{\kappa}{2} w^2$$

Without fee ($f = 0$): $w^* = c/\kappa$, clipped to $[0, \bar{w}]$.

With a fee f per traded notional: moving away from u costs $f|w - u|$, so the score must beat the fee before anything moves:

$$w^* = \begin{cases} \min\{(c - f)/\kappa, \bar{w}\} & c - \kappa u > f \text{ (buy)} \\ \max\{(c + f)/\kappa, 0\} & c - \kappa u < -f \text{ (sell)} \\ u & |c - \kappa u| \leq f \text{ (no trade)} \end{cases}$$

- the position is linear in the score inside the *position window* $0 \leq c \leq \bar{w}\kappa$: all cash below 0, fully invested above κ
- $\lambda = 1$, weekly, $\sigma = 1\%$ a day:
 $\kappa = 2 \cdot 1 \cdot 5 \cdot (1\%)^2 = 0.1\%$ per 5 days

The no-trade band is $|c - \kappa u| \leq f$: half-width $f = 3$ bp around κu , the score that would justify the position already held. A score inside it leaves the portfolio as it is, whatever it holds.

decision rate	window κ	band half-width f	band / window
weekly ($h = 5$)	0.10%	0.03%	30%
twice a week ($h = 2$)	0.04%	0.03%	75%

v1: leaf scores of ± 0.3 – 0.5% per 5 days sit far outside the window $\Rightarrow w \in \{0, 1\}$; continuous sizing needs $\lambda \geq 5$. But a score near 0 with a drifted u sits inside the band, and the tree then holds that mix (E10).

The loss: decision regret; the score of a leaf

$$U_t(w) = r_t^\top w - f^\top |w - u_t| - \lambda w^\top \Sigma_t w$$

$$\text{Reg}_t(c) = U_t(w^*(r_t; u_t, \Sigma_t)) - U_t(w^*(c; u_t, \Sigma_t)) \geq 0$$

hindsight decision – decision on c (given the same holdings u_t)

$$\hat{c}_\ell = \arg \min_{c \in [-b, b]^n} \sum_{t \in \ell} \text{Reg}_t(c), \quad b = 10\% \text{ per holding period, needed for bound + resolution}$$

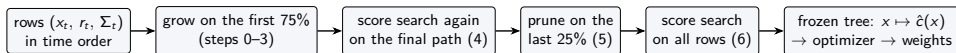
Leaf score

- one regret per date (r_t depends on t), summed over the leaf's dates
- bounded coordinate search: 3 passes $\times$ 7 values
- not a forecast: same regret for any score on the right side

Consequences

- fees $\Rightarrow$ regret depends on $u_t \Rightarrow$ holdings are frozen for the round (no backpropagation) then replayed (drifted)
- only differences: relative returns, asset vs cash $\Rightarrow$ help to choose the right features
- same Σ_t as the optimizer: unseen risk not learnt

Training: replay, two growth gates, out of sample pruning



0. root score from $u_t = 1/n$; **replay** (trade the scores date by date from u_1) $\Rightarrow u_t$, Sharpe S ; holdings frozen for the round
1. **regret gate**, per leaf: score search again from current u_t ; candidates; children of the 3 best; keep iff $\Delta/N_\ell > 10^{-6}$, $\Delta = L_\ell - L_{\text{left}} - L_{\text{right}}$ (regret per date, in h -day return)
2. **Sharpe gate**: rank all kept splits by Δ ; whole-tree replay of the first; accept iff $S' > S$, else next; one per round; none $\Rightarrow$ stop
3. replay the new tree $\Rightarrow$ new u_t ; back to 1 (depth ≤ 3 ; ≥ 20 rows and $\geq 10\%$ of the parent per leaf)
4. score search again on every leaf, final path; kept iff S does not fall
5. **pruning**, bottom-up, on the last 25% (never used for growth): with vs without the split, both replayed; $g_t = \text{Reg}_t(\text{without}) - \text{Reg}_t(\text{with})$; keep iff $\bar{g} > 10^{-6}$, $\bar{g} > k_p \text{se}(\bar{g})$ [$k_p = 2$ below the root, 0 at the root], checking Sharpe $\uparrow$
6. score search again on all training rows

Gates 1–2: in-sample. Gate 5: out-of-sample.

Split-search options: quantile candidates, refinement, soft splits

Common: admissible candidates $\theta_{(1)} < \dots < \theta_{(K)}$ (each side $\geq m_* = \max(m, \lceil \varphi N \rceil)$) rows; $m = 20$, $\varphi = 10\%$; cheap screen

$$Q(\theta) = \sum_{t \in I} \text{Reg}_t(c_{\text{side}(t)}(\theta)), \quad c_{\text{left}}(\theta) = \text{clip}(\text{mean}\{r_t : x_t \leq \theta\}, \pm b), \text{ cf Theorem 1, idem right}$$

Deciles by rank + refinement

- keep ranks $\text{round}(q(K-1))$, $q = 0.1, \dots, 0.9$: 9 screens instead of $K \approx N - 2m_*$
- refinement: screen every candidate between the two deciles around the best one, θ_o ; best of them $\theta_\bullet$
- replace iff $Q(\theta_o) - Q(\theta_\bullet) > k \text{ se}$, $\text{se} = \sqrt{|I|} \text{std}_t(\delta_t)$, $k = 1$
- 3 best survivors ranked by Q ; replay the first, then the second, then the third; keep iff Sharpe $\uparrow$

Soft split (off in v1: $\kappa = 0$)

$$z_k = \frac{Q(\theta_k) - \min_I Q(\theta_I)}{\text{se}_k}, \quad \pi_k = \frac{e^{-z_k/\kappa}}{\sum_I e^{-z_I/\kappa}}$$

- all candidates kept, weighted; Gaussian variant $e^{-z^2/2\kappa^2}$
- membership: left with $\mu(x) = \sum_{k: \theta_k \geq x} \pi_k$, right with $1 - \mu(x)$; weights $\omega_{t\ell}$ multiply down the tree
- leaf fit: $\min_c \sum_t \omega_{t\ell} \text{Reg}_t(c)$; row score $\hat{c}(x_t) = \sum_\ell \omega_{t\ell} \hat{c}_\ell$: a ramp, not a jump
- $\kappa \rightarrow 0$: plain split

Model v1: the frozen configuration (24 September 2026)

Choice	v1	Why (section of the experiments report)
risk aversion λ	1 (weight of the h -day variance)	positions 0/1: regime classifier, not sizer (8)
fee f	3 bp = 0.03% of the traded notional on the asset; 0 on cash	market-sized; 10 bp starves fast rates (8)
decision rate h	weekly ($h = 5$ trading days); every 2 days as alternative	daily $\approx$ weekly; monthly loses (4)
depth	≤ 3	depth alone changes nothing (8)
leaf minimum	20 decisions and 10% of the node	tail leaves at 0.7% of the data (7)
candidate thresholds	deciles by rank + refinement (1 s.e.)	= base Sharpe, steadier thresholds (9, 12)
soft splits	off	never raise the Sharpe at $\lambda = 1$ (9, 12)
growth gates	regret reduction, then replayed Sharpe $\uparrow$	kept as designed; v2 revisits
pruning	last 25%; hurdle 2 s.e. below the root, none at the root	out-of-sample judge (7); removes spurious leaves at no cost (12)
covariance	$\Sigma_t = h\hat{\sigma}_t^2, \hat{\sigma}_t^2$ the variance of the last 180 daily returns	20-day window: noisier than timely (regime report)
conventions	cash column; all-cash Sharpe = 0; $ \hat{c} \leq 10\%$ per holding period	

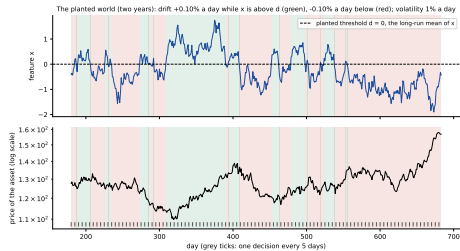
Base of every experiment = same pipeline, every option off: depth 2, all candidates screened, plain pruning.

The test-bed: a market with a planted answer (constant volatility)

$$dX_t = k(\mu - x_t)dt + \sigma dW_t, \quad dW_t \sim \mathcal{N}(0, dt)$$

$$\frac{dS_t}{S_t} = \mu(X_t)dt + \sigma dW_t, \quad \mu(X_t) = \pm e\sigma \text{ as } x_t \gtrless d$$

σ (daily)	1% (16% a year), both regimes
$e = \mu/\sigma$ (daily)	0.10, 0.05, 0.02: drift ± 25 , ± 13 , $\pm 5\%$ a year; Sharpe of the regime $e\sqrt{252} = 1.6, 0.8, 0.3$
k (daily)	0.02: half-life $\ln 2/k = 35$ days; 0.01 to 0.10 (69 to 7 days)
m, s^2, d	0, 1, 0: x standardized (mean 0, sd 1), threshold at its mean
h (trading days)	1, 2, 5, 10, 21: daily to monthly
data	daily steps (t in trading days); 15 years of training (3,780 days), 5 of test (1,260)

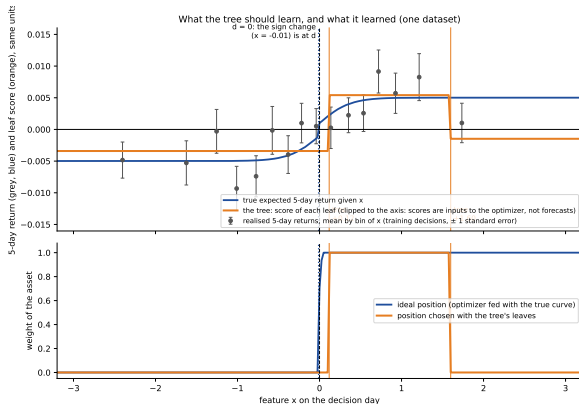


Measured on 40–100 datasets per cell (Monte Carlo)

- first threshold vs planted $d = 0$ (x has sd 1)
- annualized test Sharpe: tree vs ideal rule ($\mathbb{E}[r | x]$ fed to the optimizer), rule at d , always the asset
- share of the ideal gain captured

Second market (volatility follows a threshold too): own report; one line at the end.

One fitted tree (weekly, edge 0.10)



What is drawn (x on the decision day)

- **blue**: the true expected 5-day return given x ; grey: the realised returns, averaged by bin of x
- **orange**: the tree's score, one value per leaf, in the same units
- **bottom**: the position each one implies

Key points

- **The root split lands on the planted threshold.**
- **Scores are not forecasts.** They sit far outside the optimizer's narrow window, so the positions match the ideal rule wherever their sign is right.
- **The far-right leaf is spurious.** A tail of x sent to cash while the ideal rule stays invested; v1's pruning hurdle removes such leaves.
- **At $\lambda = 1$ the ideal rule is a step too.** The threshold is the whole problem.

Result 1: the threshold is recovered at market sizes (E2, base pipeline)

edge	h	split		threshold within 0.25	annual. test Sharpe	
		on x	(%)		tree	ideal gain
0.10	1	97	95%	0.80	1.03	76%
	5	94	81%	0.73	0.88	79%
	21	81	60%	0.54	0.68	75%
0.05	1	82	60%	0.12	0.49	25%
	5	78	53%	0.24	0.43	51%
	21	62	34%	0.17	0.35	43%
0.02	5	54	26%	-0.03	0.15	-3%

- row = 100 datasets (50 at edge 0.02), threshold planted at 0; edge e = daily drift / daily volatility: 0.10, 0.05, 0.02 = $\pm 25\%$, $\pm 13\%$, $\pm 5\%$ a year; h in trading days
- split on x : trees keeping a split (else: no split at all)
- within 0.25: first threshold within 0.25 sd of x from 0
- Sharpe: mean over datasets, s.e. ≈ 0.04
- gain: tree's excess test return over the no-split tree, as a share of the ideal rule's

- $\pm 25\%/yr$: **found at every rate**; 95% within 0.25 daily, 60% monthly; 75–79% of the ideal gain
- $\pm 13\%/yr$: **graceful degradation**; 62–82% split; 25–51% of a small gain (ideal 0.35–0.49)
- $\pm 5\%/yr$: **nothing to find in 15 years**; half no split, half at random; ideal rule ≈ 0.15 ; finding nothing = right answer
- daily $\approx$ weekly; monthly loses (regime changes inside the holding period)

Result 2: the strategy's performance out of sample (E2, base pipeline)

edge	h	annualized test Sharpe ratio				gain
		tree	ideal rule	rule at d	buy & hold	captured
0.10	1	0.80	1.03	1.04	-0.02	76%
	2	0.86	0.97	0.98	-0.02	85%
	5	0.73	0.88	0.88	-0.02	79%
	10	0.63	0.78	0.78	-0.02	78%
	21	0.54	0.68	0.66	-0.02	75%
0.05	1	0.12	0.49	0.49	-0.02	25%
	2	0.21	0.46	0.46	-0.02	44%
	5	0.24	0.43	0.42	-0.02	51%
	10	0.17	0.39	0.38	-0.02	43%
	21	0.17	0.35	0.34	-0.02	43%

- 100 datasets per row; s.e. ≈ 0.04
- ideal rule: optimizer fed $\mathbb{E}[r \mid x]$; rule at d : asset iff $x \geq d$;
buy & hold: always fully in the asset, never in cash
- turnover per weekly decision (edge 0.10, both legs): tree 0.25,
ideal 0.29 $\Rightarrow$ fees $\approx 0.2\%$ a year (3 bp on the asset leg)

- **The tree captures most of what the signal offers.** With a $\pm 25\%$ regime it earns about three quarters of the ideal rule's gain.
- **Learning the threshold is the whole job.** A rule that only knows the planted threshold matches the ideal rule: at $\lambda = 1$ there is nothing else to learn.
- **What is lost is precision.** The gap to the ideal rule is the threshold error plus a few spurious leaves.
- **A weak regime is a limit of the data, not of the tree.** At $\pm 13\%$ a year the tree keeps only part of a small gain: 15 years barely identify it.
- **Without the signal there is nothing to earn.** The asset drifts up in one regime and down in the other, so buying and holding earns zero: all the gain comes from switching with x .

Trading rate against the speed of the feature (edge 0.10, 50 datasets per cell)

half-life of x	daily		weekly		monthly	
	tree	ideal	tree	ideal	tree	ideal
69 days ($k = 0.01$)	0.63	1.09	0.84	0.97	0.68	0.84
35 days ($k = 0.02$)	0.80	1.03	0.73	0.88	0.54	0.68
14 days ($k = 0.05$)	0.69	0.98	0.65	0.75	0.30	0.49
7 days ($k = 0.10$)	0.77	0.96	0.45	0.60	0.18	0.34

annualized test Sharpe ratios, tree and ideal rule

- daily $\approx$ weekly: more decisions = more fees, not more information
- monthly loses, more so for a fast feature:
half-life 7 d $\Rightarrow$ 0.18 vs 0.34
- slow feature + daily decisions: over-trading
(0.63 vs 1.09)
- $\Rightarrow$ weekly in v1

What the design options bought (E3–E5, E8; 40 datasets per cell, edge 0.10)

setting	annual. Sharpe, tree	
	weekly	2-day
base (every option off)	0.80	0.87
depth 3	0.80	0.89
$\lambda = 5$ / $\lambda = 10$	0.70 / 0.65	0.81 / 0.71
fee 10 bp	0.60	0.53
deciles + refinement	0.82	0.82
soft splits $\kappa = 1$	0.53	0.35
soft splits $\kappa = 0.25$	0.73	0.61
hurdle 2 s.e., every node	0.43	0.47
hurdle 2 s.e., below the root only	0.77	0.74
deciles + refin. + hurdle below root	0.80	0.79

base = the tree and training of Results 1–2 with every option off: depth ≤ 2 , every threshold screened, pruning without a hurdle, $\lambda = 1$, fee 3 bp. Each other row changes one thing from it. Ideal rule 0.91 weekly, 1.00 at 2 days; s.e. ≈ 0.04 ; last row = v1 at depth 2

- **No option beats the base beyond noise.** Every difference sits inside two standard errors; what remains is the noise of the threshold itself.
- **Depth, risk aversion and fees do what they should.** Depth 3 changes nothing, one threshold needs one split; $\lambda = 5$ –10 gives gradual positions at a cost of 0.1 to 0.15; a 10 bp fee stops the fast rates from trading.
- **Deciles with refinement, kept in v1.** Same Sharpe as screening every threshold, with 9 candidates instead of hundreds and steadier thresholds.
- **A pruning hurdle at every node, root included: not kept.** Asking the root split to beat two standard errors on the checking rows removes true root splits: the last quarter of the training period is too short to certify a split at that confidence.
- **The hurdle below the root only, kept in v1.** It removes the spurious second-level leaves, from 25% of the trees to 5% at twice a week and to none from weekly on, at no cost in Sharpe from weekly on.

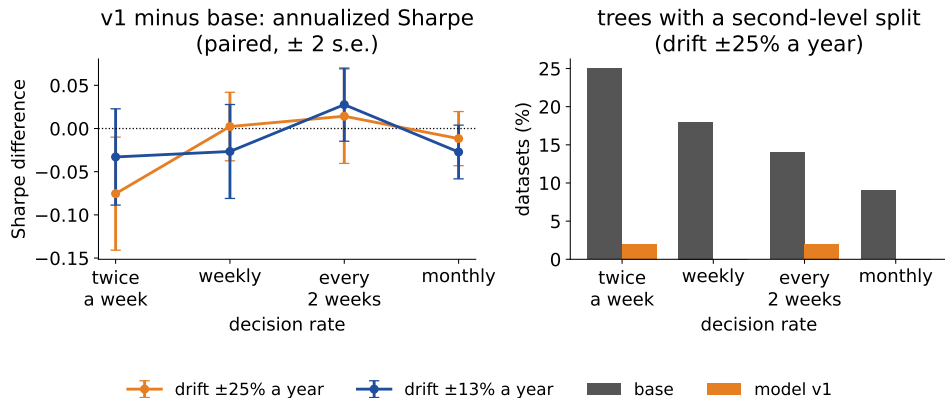
Model v1 against the base, on the same 100 datasets (E9, 800 fits)

edge	h	leaves		2nd level	within 0.25	Sharpe		v1 — base (2 s.e.)
		base	v1	base → v1	base → v1	base	v1	
0.10	2	2.24	2.00	25 → 2%	92 → 89%	0.86	0.78	−0.08 (0.07)
	5	2.14	1.94	18 → 0%	81 → 82%	0.73	0.73	+0.00 (0.04)
	10	2.03	1.92	14 → 2%	70 → 72%	0.63	0.64	+0.01 (0.05)
	21	1.90	1.81	9 → 0%	60 → 64%	0.54	0.53	−0.01 (0.03)
0.05	2	2.18	1.72	38 → 4%	64 → 57%	0.21	0.18	−0.03 (0.06)
	5	2.01	1.80	21 → 3%	53 → 51%	0.24	0.22	−0.03 (0.05)
	10	1.93	1.69	19 → 1%	47 → 49%	0.17	0.20	+0.03 (0.04)
	21	1.76	1.59	13 → 0%	34 → 42%	0.17	0.14	−0.03 (0.03)

same 100 datasets (seeds 0–99); Sharpe: annualized test Sharpe ratio;
 2nd level: datasets keeping a second-level split; last column: mean paired
 difference of annualized test Sharpe, 2 s.e. in brackets; half of the
 datasets: identical Sharpe

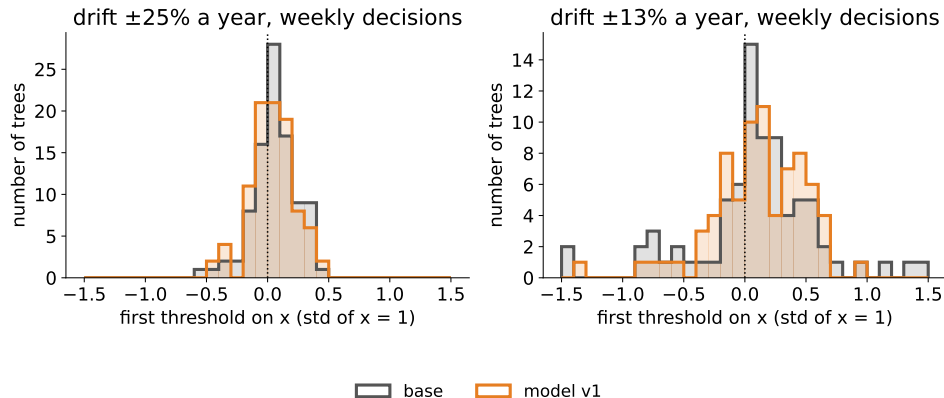
- **The question.** Does the frozen v1 do better than the base it was built from? Both are fitted on the same 100 histories, so every difference is paired.
- **Same Sharpe.** From weekly decisions on, v1 and the base are within noise of each other.
- **Cleaner trees.** The spurious second-level leaves are gone, the threshold is found as often, and the trees are smaller.
- **One cost.** Deciding every 2 days, v1 loses a little: some second-level splits are real there, and the pruning hurdle removes them.
- **Verdict: v1 is the base with a cleaner structure, not a higher Sharpe.**

Model v1 against the base: the paired differences in one look (E9)



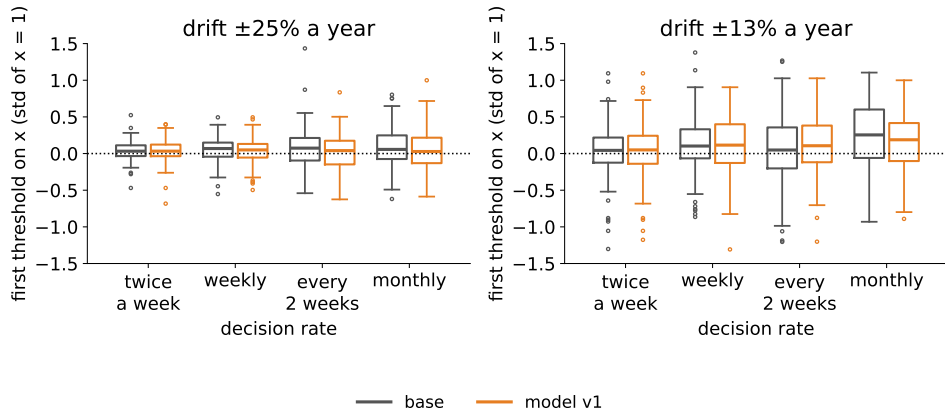
Same 100 datasets for v1 and the base (seeds 0–99), 15 years of training, 5 of test. Left: annualized test Sharpe ratio of v1 minus that of the base, dataset by dataset; mean and two standard errors. Right: share of the datasets whose tree keeps a second-level split, at a drift of $\pm 25\%$ a year.

E9: the distribution of the first threshold, v_1 against the base



Same 100 datasets per panel, weekly decisions, $\lambda = 1$, fee 3 bp. Every tree that splits on x contributes its first threshold; the planted value is 0. Model v1 in orange, the base in grey.

E9: the first threshold by decision rate, v1 against the base



E9, $\lambda = 1$, fee 3 bp, 100 datasets per box. Boxes: median and quartiles over the trees that split on x ; whiskers at 1.5 interquartile ranges, dots beyond. The planted value is 0.

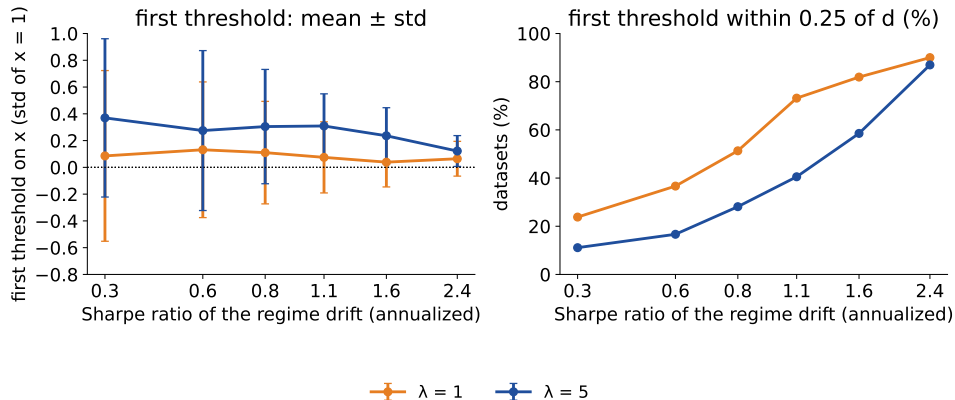
What can be found at all: resolution against volatility and feature noise (E6)

sweep (weekly)	value	edge	median $ \theta $	tree / ideal
σ per day (a year)	0.4% (6%)	0.25	0.05	1.85 / 2.10
	1% (16%)	0.10	0.12	0.69 / 0.87
	2% (32%)	0.05	0.39	0.23 / 0.42
	4% (63%)	0.025	0.95	-0.05 / 0.18
$x + \tau \varepsilon$ observed	0	0.10	0.12	0.69 / 0.87
	0.25	0.10	0.19	0.66 / 0.79
	0.5	0.10	0.25	0.45 / 0.66
	1	0.10	0.55	0.26 / 0.48
	2	0.10	1.15	0.09 / 0.29

60 datasets per row; volatility sweep at a fixed drift of $\pm 25\%$ /yr;
 $|\theta|$: first threshold's distance from the planted 0 (sd of x);
 $\varepsilon \sim \mathcal{N}(0, 1)$ daily white noise, τ its std in std of x ($= 1$); the
price follows the true x ; tree / ideal: annualized test Sharpe

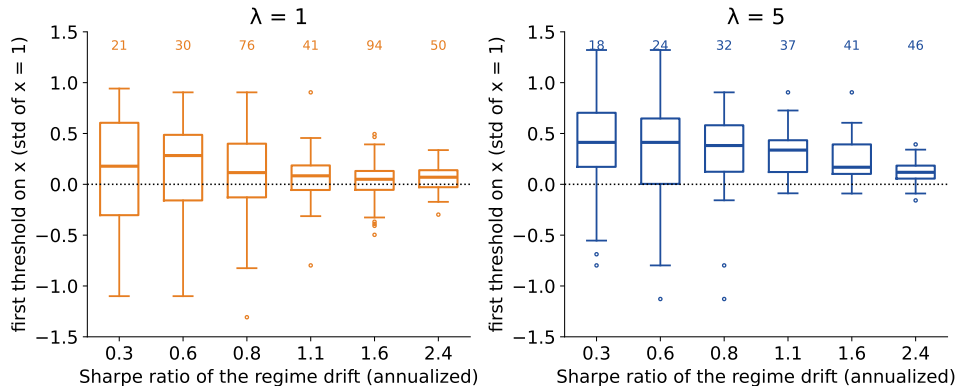
- threshold error $\approx 0.012/e$ down to $e \approx 0.05$; below: a guess
- noise τ on $x \approx$ edge divided by $1 + \tau^2$
- $\tau \leq 0.25$: free; 0.5: error doubles; 1: ideal Sharpe halves
- edge below a few bp/day: unlearnable from 15 years, by any method
- $\Rightarrow$ pre-test every real feature with this protocol

E10: the planted threshold against the strength of the signal



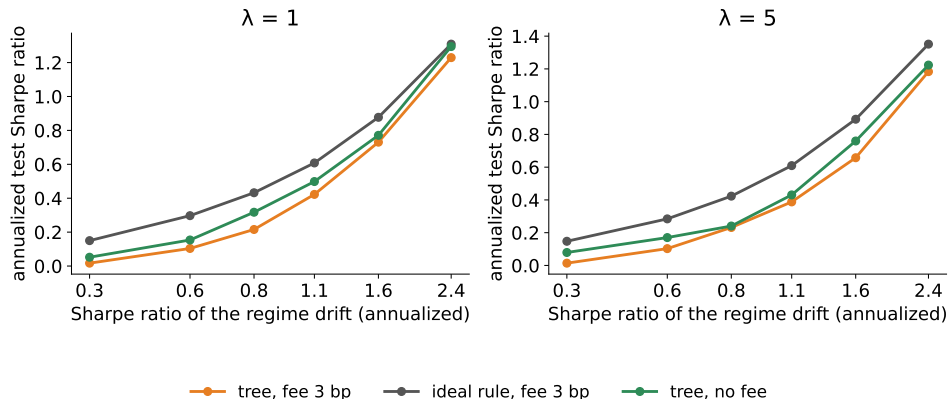
Model v1, weekly decisions, fee 3 bp, 16% volatility, half-life of x 35 days. The drift goes from ± 5 to $\pm 38\%$ a year, that is a Sharpe ratio of the regime drift from 0.3 to 2.4; 50 datasets per point (100 at 0.8 and 1.6). Left: the first threshold found, mean and one standard deviation over the datasets that split; the planted value is 0. Right: share of the datasets whose first threshold is within 0.25 of it.

E10: the distribution of the first threshold by signal strength



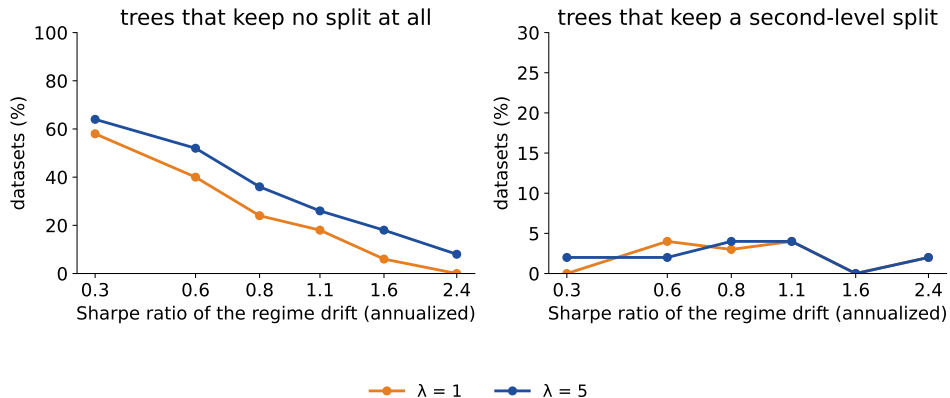
E10, weekly decisions, fee 3 bp, 16% volatility, half-life 35 days. The number above each box is how many of the 50 datasets (100 at 0.8 and 1.6) kept a split on x . Boxes: median and quartiles; whiskers at 1.5 interquartile ranges; dots beyond. At $\lambda = 5$ the whole distribution sits above the planted value.

E10: the Sharpe ratio against the strength of the signal, with and without fees



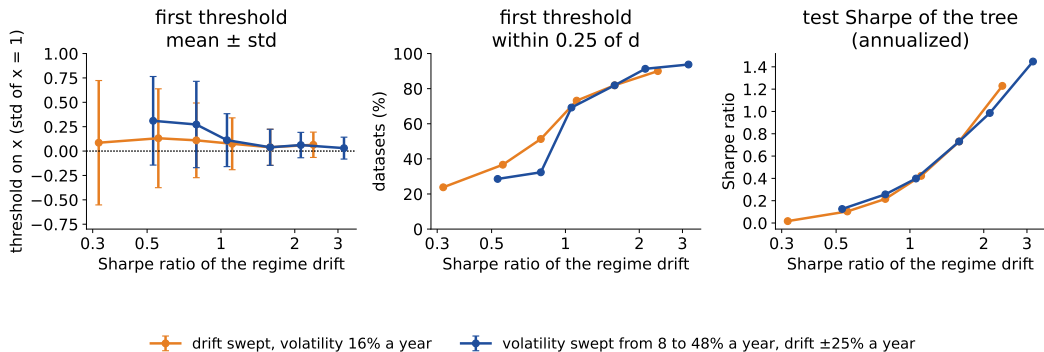
Same sweep. Annualized test Sharpe ratio of the tree with the 3 bp fee, of the tree without fees, and of the ideal rule (the optimizer fed with the true expected return), at $\lambda = 1$ and $\lambda = 5$.

E10: trees that keep no split, trees that keep a second-level split



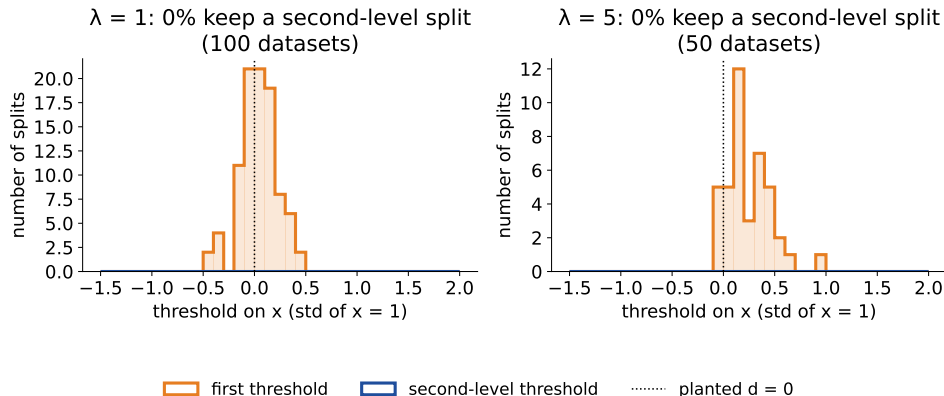
Same sweep, fee 3 bp. Left: share of the datasets where pruning leaves a single leaf. Right: share where a split below the root survives; the market has one threshold, so such splits are spurious.

E10: does only the Sharpe ratio matter? Volatility sweep against drift sweep



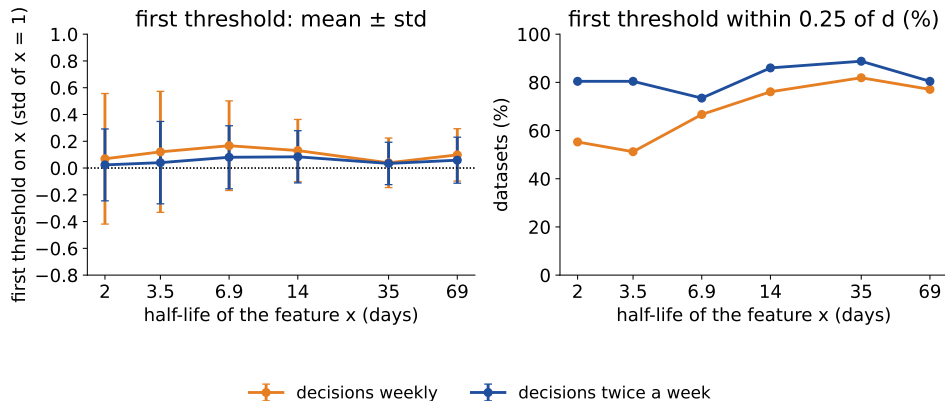
Model v1, $\lambda = 1$, weekly, fee 3 bp, half-life 35 days, 50 datasets per point. Orange: the sweep of frame 22, the drift raised at a fixed 16% volatility. Blue: the volatility raised from 8 to 48% a year at a fixed drift of $\pm 25\%$, which lowers the same Sharpe ratio from 3.2 (8%) to 0.5 (48%). **Why this frame:** if the Sharpe ratio of the drift were all that counted, the two curves would coincide. They do down to 24% volatility; at 32–48% the blue points fall below, because the risk term $\kappa = 2\lambda\sigma^2$ becomes as large as the drift, the ideal position turns into a ramp, and the threshold shifts up, as it does at $\lambda = 5$.

E10: where the second-level thresholds land



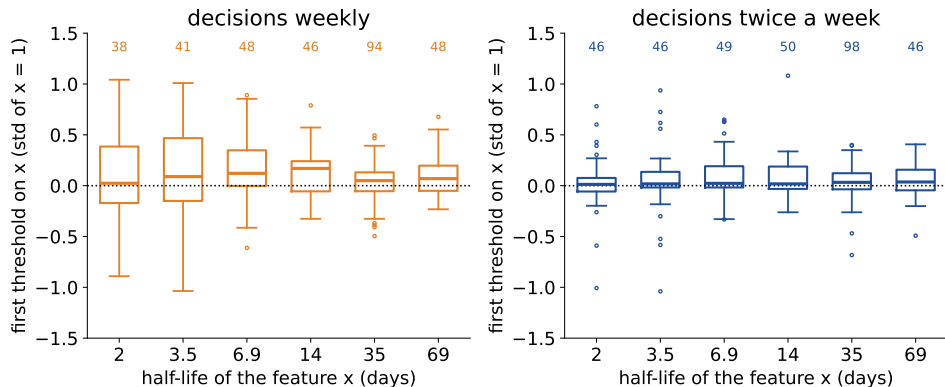
Model v1, weekly, drift $\pm 25\%$ a year, 16% volatility, fee 3 bp. Every threshold the trees keep: the first thresholds in orange, the second-level ones in blue. There are none: v1's pruning hurdle removes every split below the root in this market.

E10: the half-life of the feature: the threshold



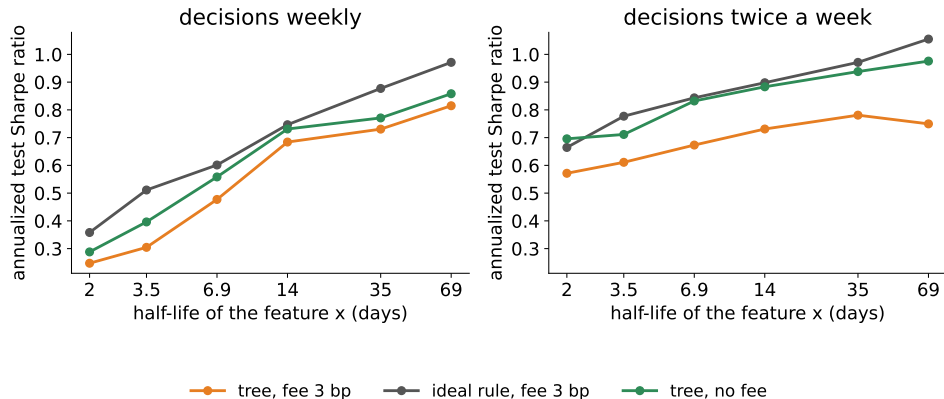
Model v1, $\lambda = 1$, fee 3 bp, drift $\pm 25\%$ a year, 16% volatility, 50 datasets per point. The half-life of x goes from 2 to 69 days (35 days everywhere else in this deck); decisions weekly or twice a week.

E10: the distribution of the first threshold by half-life



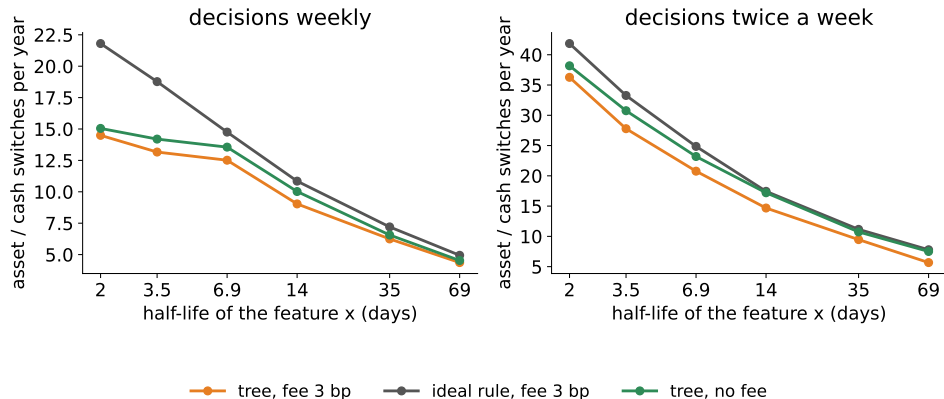
E10, $\lambda = 1$, fee 3 bp, drift $\pm 25\%$ a year, 16% volatility. The number above each box is how many of the 50 datasets (100 at 35 days) kept a split on x . Boxes: median and quartiles; whiskers at 1.5 interquartile ranges; dots beyond.

E10: the half-life of the feature: the Sharpe ratio



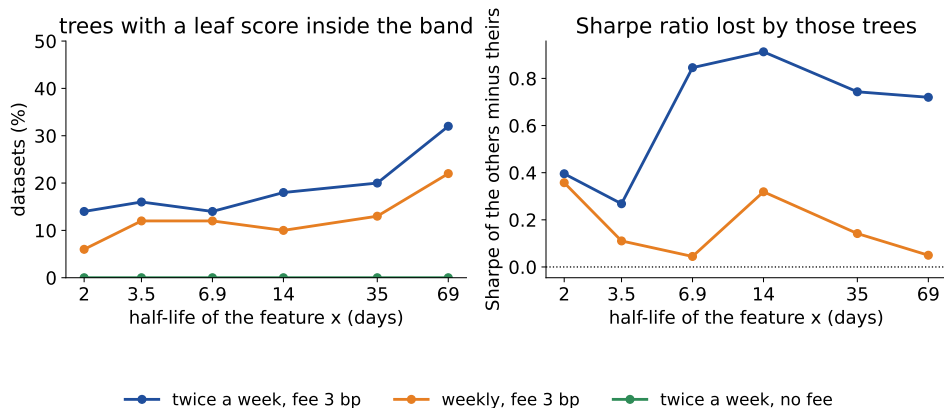
Same sweep. Annualized test Sharpe ratio of the tree with the 3 bp fee, of the tree without fees, and of the ideal rule.

E10: the half-life of the feature: how often the tree switches



Same sweep. Number of full moves between the asset and cash per year on the test period, tree and ideal rule. At 3 bp a full move costs 0.03%, so 36 a year cost about 0.1% a year.

E10: the trees that freeze inside the no-trade band



Model v1, $\lambda = 1$, drift $\pm 25\%$ a year, 16% volatility, 50 datasets per point. With a 3 bp fee the optimizer does not trade while the score stays within $\pm 0.03\%$ of what it holds; twice a week that band is as wide as the whole position window. Left: share of trees with a leaf score inside the band. Right: the annualized test Sharpe ratio of the other trees minus that of the frozen ones, same series and colours; the no-fee series has no frozen tree, so nothing to draw.

E10 in six points

- **The Sharpe ratio of the regime drift sets the precision.** A $\pm 25\%$ regime is found within 0.25 of the threshold in 8 datasets out of 10, a $\pm 5\%$ one in 1 out of 4, and then half the trees keep no split. Fees change almost nothing.
- **Volatility acts through that Sharpe ratio**, until 32–48% a year, where the risk term makes the ideal position gradual and the recovery drops below what the Sharpe ratio alone predicts.
- **At $\lambda = 5$ the split sits above the threshold, on purpose.** The ideal position is a ramp in x ; the best single split cuts it in the middle, not at d .
- **Second splits are gone.** At most 8% of the datasets keep one.
- **A fast feature needs a fast decision rate, and the fee bill is not the limit.** With a half-life of a few days, weekly decisions cannot follow the regime; the tree switches less often than the ideal rule, not more. Deciding twice a week recovers the threshold and most of the ideal Sharpe at about 0.1% a year of fees.
- **But the fee inside the training freezes one tree in five at twice a week.** A leaf score that lands within 0.03% of zero sits in the optimizer's no-trade band, and the tree then holds whatever it has, a drifted mix, for years: those trees earn 0.2 instead of 0.9. Without the fee in the training none freezes. This is the case for taking the fees out of the regret (v2, item 1).

E11: the second market: a threshold for the drift and one for the volatility

The feature is the same daily Ornstein–Uhlenbeck process; the price now has two regimes for its drift and two for its volatility (t in trading days):

$$x_{t+1} = m + e^{-k}(x_t - m) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, s^2(1 - e^{-2k}))$$
$$\log \frac{S_{t+1}}{S_t} = \mu_t - \frac{1}{2}\sigma_t^2 + \sigma_t \varepsilon_t, \quad \mu_t = \begin{cases} +\mu & x_t \geq d \\ -\mu & x_t < d \end{cases}, \quad \sigma_t = \begin{cases} \rho\sigma & x_t \geq d_\sigma \\ \sigma & x_t < d_\sigma \end{cases}$$

$\sigma = 1\%$ a day (16% a year), $\rho = 0.5$ (the calm regime at half the volatility), $\mu = e\sigma$ with $e = 0.10$ (drift $\pm 25\%$ a year), $d = 0$; everything else as in the first market.

Two placements of the volatility threshold

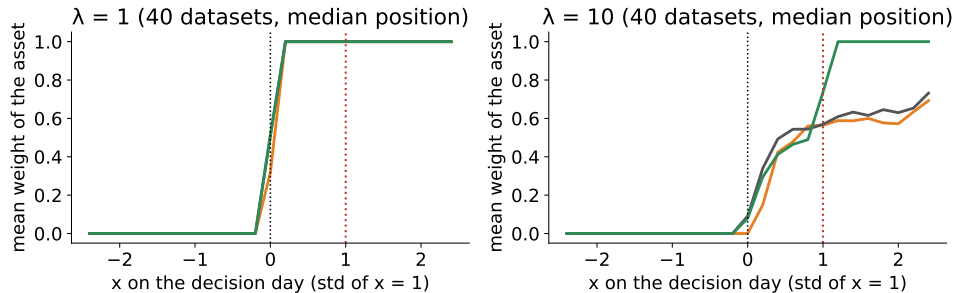
- $d_\sigma = d$: the bad regime is the noisy one, the usual picture of markets
- $d_\sigma = d + 1$: the good regime is split into a stressed part, $0 \leq x < 1$, and a calm part, $x \geq 1$

What a correct tree should do

- the expected return given x does not depend on the volatility: the drift threshold stays at $d = 0$
- the volatility acts through the position, $w = (g_h(x) \mp f)/(2\lambda\sigma_h^2)$ clipped to $[0, 1]$: at $\lambda = 1$ all or nothing in both regimes, so the volatility threshold is invisible; at $\lambda = 10$ about one half in the stressed good regime and one in the calm one, a second boundary at $x = 1$ of size rather than of sign

Seven settings: constant volatility, $d_\sigma = d$, $d_\sigma = d + 1$, each at $\lambda = 1$ and $\lambda = 10$, plus $d_\sigma = d + 1$ at $\lambda = 10$ with a 20-day covariance; decisions every 2, 5, 10 and 21 days, drifts ± 13 and $\pm 25\%$ a year, 40 datasets per cell: 2,240 fits, on the base in E7 (23 September) and on model v1 in E11 (28 September).

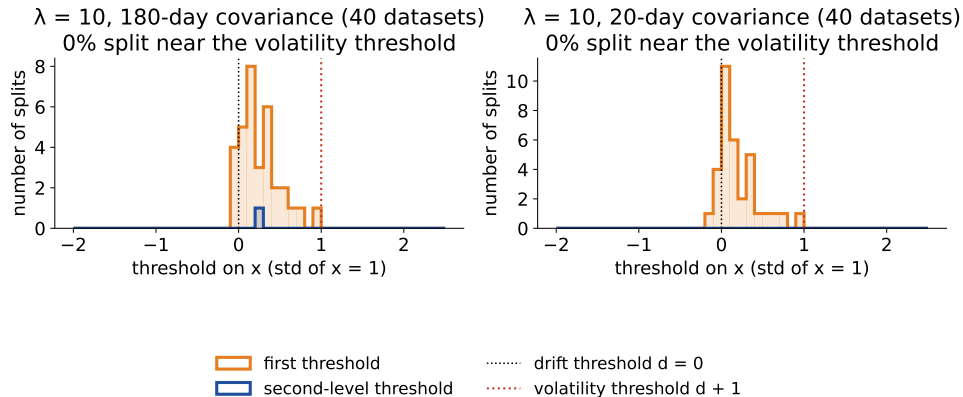
E11: the second market on model v1: the position taken by x



— tree
— ideal rule, same lagged covariance
— ideal rule, true variance
..... drift threshold $d = 0$
..... volatility threshold $d + 1$

E7 rerun on model v1 (E11, 2,240 fits): volatility 1% a day below $x = 1$ and 0.5% above, drift threshold at 0; weekly, drift $\pm 25\%$ a year, 40 datasets; median position by x on the test decisions. At $\lambda = 1$ the position is 0 or 1 whatever the variance. At $\lambda = 10$ the ideal position is one half in the stressed part of the good regime and one in the calm part; the tree follows the ideal rule fed with the same lagged covariance, not the one that knows the true variance.

E11: the second market on model v1: no split at the volatility threshold



Same setting, $\lambda = 10$: every threshold kept by the 40 trees, with the 180-day covariance and with a 20-day one that follows the regime. Model v1 keeps no split near the volatility threshold at all, where the base kept one in 8% of the datasets. The boundary of the variance is invisible to the regret, which uses the same lagged covariance as the optimizer, and the pruning hurdle removes the few splits the base had found there.

A fair assessment, and what comes next

Holds

- $\pm 25\%$ regime, monthly feature: recovered at every rate; 75–85% of the attainable gain
- $\pm 13\%$: graceful; $\pm 5\%$: correctly nothing
- v1 = base with smaller trees, steadier thresholds (100 paired datasets); cost 0.08 at 2 days
- second market (E7 on the base, E11 on v1): drift threshold found better when the calm regime pays, Sharpes nearly double; a variance-only boundary is never learnt, by v1 not once (lagged covariance in the regret)

Weak or untested

- no option beats the base beyond noise: structure, not Sharpe
- growth Sharpe gate: in-sample, mixes a risk preference into the structure
- one clean feature; synthetic data only; decoys only at an unrealistic edge; depth budget unused

Next (v2)

- growth gate off / replayed regret instead; same 100 datasets
- regime-aware variance for optimizer and regret
- two features + decoys at market-sized edges
- real data; this protocol as the pre-test of every feature